Database Systems

Lecture #07

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Objectives

- To learn relational algebra
  - Operators of relational algebra
  - Query expressions with relational algebra

Outline

- Relational Algebra
- SELECT Operation
- PROJECT Operation
- Relational Algebra Expressions
- Set Operations
- CARTESIAN PRODUCT Operation
- JOIN Operation
- A Complete Set of Relational Algebra Operations

Relational Algebra

- Basic set of *operations* for the relational model
- Used for specifying *retrieval requests*
- A result of a retrieval is represented as a new *relation*

Relational Algebra

- Types
  - SELECT
  - PROJECT
  - Set operations
  - JOIN
  - …

SELECT Operation

- Format

$$\sigma_{<\text{selection condition}>}(R)$$

SELECT Operation

- Meaning
  - Selects a *subset of the tuples* from a relation *R* that satisfies a selection condition *c*
  - Selection condition *c* is a Boolean expression
    - <attribute name> <comparison op> <constant value>
    - <attribute name> <comparison op> <attribute name>
  - A resulting relation contains *only* the tuples which satisfies *c* from *r*(*R*)

SELECT Operation

- Examples

$$\sigma_{Dno=4}(EMPLOYEE)$$
$$\sigma_{Salary>30000}(EMPLOYEE)$$

$$\sigma_{(Dno=4\ \textbf{AND}\ Salary>25000)\ \textbf{OR}\ (Dno=5\ \textbf{AND}\ Salary>30000)}(EMPLOYEE)$$

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5

PROJECT Operation

- Format

$$\pi_{<\text{attribute list}>}(R)$$

PROJECT Operation

- Meaning
  - *Selects columns* specified in attribute list L from table R
    - Discards the other columns
  - A resulting relation contains those tuples with attributes specified in *L* from *r*(*R*)
  - The result of PROJECT operation is a set of *distinct tuples*

PROJECT Operation

- Examples

$\pi_{\text{Lname, Fname, Salary}}(\text{EMPLOYEE})$

Lname	Fname	Salary
Smith	John	30000
Wong	Franklin	40000
Zelaya	Alicia	25000
Wallace	Jennifer	43000
Narayan	Ramesh	38000
English	Joyce	25000
Jabbar	Ahmad	25000
Borg	James	55000

PROJECT Operation

- Examples

$\pi_{Sex,\ Salary}(EMPLOYEE)$

Sex	Salary
M	30000
M	40000
F	25000
F	43000
M	38000
M	25000
M	55000

Relational Algebra Expressions

- Sequence of relational algebra operations
- Complex queries can be represented as a relational algebra expression
  - A relational algebra expression *results in a relation*

Relational Algebra Expressions

- Examples
  - "Retrieve the first name, last name, and salary of all employees who work in department number 5"
    - $\pi_{\text{Fname, Lname, Salary}}(\sigma_{\text{Dno}=5}(\text{EMPLOYEE}))$
  - Each intermediate result can have a relation name
    - $\text{DEP5_EMPS} \leftarrow \sigma_{\text{Dno}=5}(\text{EMPLOYEE})$
      $\text{RESULT} \leftarrow \pi_{\text{Fname, Lname, Salary}}(\text{DEP5_EMPS})$

Relational Algebra Expressions

- Examples
  - "Retrieve the first name, last name, and salary of all employees who work in department number 5"
    - $\pi_{\text{Fname, Lname, Salary}}(\sigma_{\text{Dno}=5}(\text{EMPLOYEE}))$
  - Each attribute of a result relation can be *renamed*
    - $\text{TEMP} \leftarrow \sigma_{\text{Dno}=5}(\text{EMPLOYEE})$
      $R(\text{First_name, Last_name, Salary}) \leftarrow \pi_{\text{Fname, Lname, Salary}}(\text{TEMP})$

Set Operations

- Relational algebra operations from set theory
  - UNION operation
    - $R_1 \cup R_2$
  - INTERSECTION operation
    - $R_1 \cap R_2$
  - SET DIFFERENCE (MINUS) operation
    - $R_1 - R_2$
  - CARTESIAN PRODUCT (CROSS PRODUCT) operation
    - $R_1 \times R_2$

Set Operations

- Example

$$DEP5_EMPS \leftarrow \sigma_{Dno=5}(EMPLOYEE)$$
$$RESULT1 \leftarrow \pi_{Ssn}(DEP5_EMPS)$$
$$RESULT2(Ssn) \leftarrow \pi_{Super_ssn}(DEP5_EMPS)$$
$$RESULT \leftarrow RESULT1 \cup RESULT2$$

RESULT1

Ssn
123456789
333445555
666884444
453453453

RESULT2

Ssn
333445555
888665555

RESULT

Ssn
123456789
333445555
666884444
453453453
888665555

Set Operations

- Union compatibility (type compatibility)
  - Two relations should have the same *type of tuples*
  - For set operations ∪, ∩, and -
    - The two relation $R(A_1, A_2, ..., A_n)$, $S(B_1, B_2, ..., B_n)$ should have
      - *Same number of attributes n*
    - Two attributes in the corresponding attribute pair *should have the domains compatible (similar) to each other*
      - $dom(A_i) = dom(B_i)$ $(1 \leq i \leq n)$

Set Operations

- Examples
  - Two union compatible relations

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

Set Operations

- Examples
  - STUDENT ∪ INSTRUCTOR

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert
John	Smith
Ricardo	Browne
Francis	Johnson

Set Operations

- Examples
  - STUDENT ∩ INSTRUCTOR

Fn	Ln
Susan	Yao
Ramesh	Shah

Set Operations

- Examples
  - STUDENT - INSTRUCTOR

Fn	Ln
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

Set Operations

- Examples
  - INSTRUCTOR - STUDENT

Fname	Lname
John	Smith
Ricardo	Browne
Francis	Johnson

CARTESIAN PRODUCT Operation

- Format

$$R \times S$$

or

$$R(A_1, A_2, \ldots, A_n) \times S(B_1, B_2, \ldots, B_m)$$

CARTESIAN PRODUCT Operation

- Meaning
  - Produces a new relation by *combining* every tuple from one relation with every tuple from the other relation
  - $Q(A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_m)$
    - Number of attributes: $n + m$
    - Number of tuples: $n_R * n_S$
  - Useful only when followed by a selection that matches values of attributes

CARTESIAN PRODUCT Operation

- Example
  - "Retrieve a list of names of each female employee's dependents"

$$\text{FEMALE_EMPS} \leftarrow \sigma_{\text{Sex}='\text{F}'}(\text{EMPLOYEE})$$
$$\text{EMPNAMES} \leftarrow \pi_{\text{Fname, Lname, Ssn}}(\text{FEMALE_EMPS})$$
$$\text{EMP_DEPENDENTS} \leftarrow \text{EMPNAMES} \times \text{DEPENDENT}$$
$$\text{ACTUAL_DEPENDENTS} \leftarrow \sigma_{\text{Ssn=Essn}}(\text{EMP_DEPENDENTS})$$
$$\text{RESULT} \leftarrow \pi_{\text{Fname, Lname, Dependent_name}}(\text{ACTUAL_DEPENDENTS})$$

CARTESIAN PRODUCT Operation

- Example

$$\text{FEMALE_EMPS} \leftarrow \sigma_{\text{Sex}='\text{F}'}(\text{EMPLOYEE})$$
$$\text{EMPNAMES} \leftarrow \pi_{\text{Fname, Lname, Ssn}}(\text{FEMALE_EMPS})$$

FEMALE_EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Alicia	J	Zelaya	999887777	1968-07-19	3321Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291Berry, Bellaire, TX	F	43000	888665555	4
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

EMPNAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

CARTESIAN PRODUCT Operation

- Example

$$EMP_DEPENDENTS \leftarrow EMPNAMES \times DEPENDENT$$

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	...
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	...
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	...
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	...
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	...
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	...
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	...
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	...
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	...
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	...
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	...
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	...

CARTESIAN PRODUCT Operation

- Example

$$\text{ACTUAL_DEPENDENTS} \leftarrow \sigma_{\text{Ssn=Essn}}(\text{EMP_DEPENDENTS})$$
$$\text{RESULT} \leftarrow \pi_{\text{Fname, Lname, Dependent_name}}(\text{ACTUAL_DEPENDENTS})$$

ACTUAL_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...

RESULT

Fname	Lname	Dependent_name
Jennifer	Wallace	Abner

JOIN Operation

- Necessity
  - Used to process *relationships* among tuples stored in different relations

JOIN Operation

- Example
  - "Retrieve the name of the manager of each department"

DEPT_MGR

Dname	Dnumber	Mgr_ssn	...	Fname	Minit	Lname	Ssn	...
Research	5	333445555	...	Franklin	T	Wong	333445555	...
Administration	4	987654321	...	Jennifer	S	Wallace	987654321	...
Headquarters	1	888665555	...	James	E	Borg	888665555	...

JOIN Operation

- Example
  - Using CARTESIAN PRODUCT
    - ALL_PRODUCT ← DEPARTMENT × EMPLOYEE
    - DEPT_MGR ← $\sigma_{Mgr_ssn=Ssn}$ALL_PRODUCT
    - RESULT ← $\pi_{Dname,\ Lname,\ Fname}$(DEPT_MGR)

JOIN Operation

- Example
  - Using JOIN

$$DEPT_MGR \leftarrow DEPARTMENT \bowtie_{Mgr_ssn=Ssn} EMPLOYEE$$
$$RESULT \leftarrow \pi_{Dname,\ Lname,\ Fname}(DEPT_MGR)$$

THETA JOIN Operation

- Format

$$R \bowtie_{\text{<join condition>}} S$$

THETA JOIN Operation

- Meaning
  - Produces a new relation by combining every tuple from one relation with every tuple from the other relation
    - Whenever the combination *satisfies the join condition c*
  - SELECT operation after CARTESIAN PRODUCT
  - Join condition *c*:
    - <condition> **AND** <condition> **AND**...**AND** <condition>
    - <condition> = $A_i \theta B_j$
    - $\theta = \{=, <, \leq, >, \geq, \neq\}$

EQUIJOIN Operation

- Meaning
  - *Only = comparison operator* used in a JOIN operation
  - The join result always has *a pair of attributes that have identical values* in every tuple

NATURAL JOIN Operation

- Format

$$R * S$$

NATURAL JOIN Operation

- Meaning
  - EQUIJOIN
    - The join result always has *a pair of attributes that have identical values* in every tuple => *Unnatural!*
  - Second (superfluous) attribute in an EQUIJOIN condition is removed in the final result
  - Before joining, two join attributes must have the same name in both relations

NATURAL JOIN Operation

- Examples
  - "Combine each PROJECT tuple with the DEPARTMENT tuple that controls the project"

$$DEPT \leftarrow \rho_{(Dname,\ Dnum,\ Mgr_ssn,\ Mgr_start_date)}(DEPARTMENT)$$
$$PROJ_DEPT \leftarrow PROJECT * DEPT$$

Makes attributes have the same name

PROJ_DEPT

Pname	Pnumber	Plocation	Dnum	Dname	Mgr_ssn	Mgr_start_date
ProductX	1	Bellaire	5	Research	333445555	1988-05-22
ProductY	2	Sugarland	5	Research	333445555	1988-05-22
ProductZ	3	Houston	5	Research	333445555	1988-05-22
Computerization	10	Stafford	4	Administration	987654321	1995-01-01
Reorganization	20	Houston	1	Headquarters	888665555	1981-06-19
Newbenefits	30	Stafford	4	Administration	987654321	1995-01-01

NATURAL JOIN Operation

- Examples
  - "Combine each DEPARTMENT tuple with its location"

$$DEPT_LOCS \leftarrow DEPARTMENT * DEPT_LOCATIONS$$

Already have same name

DEPT_LOCS

Dname	Dnumber	Mgr_ssn	Mgr_start_date	Location
Headquarters	1	888665555	1981-06-19	Houston
Administration	4	987654321	1995-01-01	Stafford
Research	5	333445555	1988-05-22	Bellaire
Research	5	333445555	1988-05-22	Sugarland
Research	5	333445555	1988-05-22	Houston

JOIN Operation

- Two relations can be joined with more than one JOIN attribute
  - With different meanings

JOIN Operation

- Examples
  - "Associate each DEPARTMENT with its manager"

$$\text{DEPARTMENT} \bowtie_{\text{Mgr_ssn=Ssn}} \text{EMPLOYEE}$$

  - "Associate each EMPLOYEE with the department for which the EMPLOYEE works"

$$\text{EMPLOYEE} \bowtie_{\text{Dno=Dnumber}} \text{DEPARTMENT}$$

JOIN Operation

- A single relation can be used in both sides of JOIN operation
  - Treated as a JOIN between two identical copies of the same relation
  - Renaming is useful

JOIN Operation

- Examples
  - "Retrieve the name of each EMPLOYEE and the name of its supervisor"

    SUPERVISOR(Super_ssn, Sfname, Slname)

    $$\leftarrow \pi_{Ssn,\ Fname,\ Lname}(EMPLOYEE)$$

    TEMP $\leftarrow$ EMPLOYEE * SUPERVISOR

    RESULT $\leftarrow \pi_{Fname,\ Lname,\ Sfname,\ Slname}(TEMP)$

A Complete Set of Relational Algebra Operations

- *Complete set*
  - Any relational algebra operation can be expressed
  - As a sequence of operations from this set

A Complete Set of Relational Algebra Operations

- Complete set of relational algebra
  - {SELECT, PROJECT, UNION, SET DIFFERENCE, CARTESIAN PRODUCT}
  - JOIN?
    - can be represented as a SELECT after CARTESIAN PRODUCT
  - INTERSECTION?
    - Meaning?
    - How to express by the complete set?
  - DIVISION? (out of scope)
    - Meaning?
    - How to express by the complete set?

A Complete Set of Relational Algebra Operations

- *Relationally complete* languages
  - Query languages capable of express *complete set of relational algebra operations*

Summary

OPERATION	PURPOSE	NOTATION
SELECT	Selects all tuples that satisfy the selection condition from a relation R.	$\sigma_{\text{<selection condition>}}(R)$
PROJECT	Produces a new relation with only some of the attributes of R, and removes duplicate tuples.	$\pi_{\text{<attribute list>}}(R)$
THETA JOIN	Produces all combinations of tuples from R_1 and R_2 that satisfy the join condition.	$R_1 \bowtie_{\text{<join condition>}} R_2$
EQUIJOIN	Produces all the combinations of tuples from R_1 and R_2 that satisfy a join condition with only equality comparisons.	$R_1 \bowtie_{\text{<join condition>}} R_2$, OR $R_1 \bowtie_{(\text{<join attributes 1>}), (\text{<join attributes 2>})} R_2$
NATURAL JOIN	Same as EQUIJOIN except that the join attributes of R_2 are not included in the resulting relation; if the join attributes have the same names, they do not have to be specified at all.	$R_1 *_{\text{<join condition>}} R_2$, OR $R_1 *_{(\text{<join attributes 1>}), (\text{<join attributes 2>})} R_2$ OR $R_1 * R_2$

Summary

OPERATION	PURPOSE	NOTATION
UNION	Produces a relation that includes all the tuples in R_1 or R_2 or both R_1 and R_2; R_1 and R_2 must be union compatible.	$R_1 \cup R_2$
INTERSECTION	Produces a relation that includes all the tuples in both R_1 and R_2; R_1 and R_2 must be union compatible.	$R_1 \cap R_2$
DIFFERENCE	Produces a relation that includes all the tuples in R_1 that are not in R_2; R_1 and R_2 must be union compatible.	$R_1 - R_2$
CARTESIAN PRODUCT	Produces a relation that has the attributes of R_1 and R_2 and includes as tuples all possible combinations of tuples from R_1 and R_2.	$R_1 \times R_2$
DIVISION	Produces a relation $R(X)$ that includes all tuples $t[X]$ in $R_1(Z)$ that appear in R_1 in combination with every tuple from $R_2(Y)$, where $Z = X \cup Y$.	$R_1(Z) \div R_2(Y)$

Summary

- Relational Algebra Operations
  - SELECT
  - PROJECT
  - Set Operations
    - UNION, INTERSECTION, SET DIFFERENCE, CARTESIAN PRODUCT
  - JOIN
    - THETA JOIN
    - EQUIJOIN
    - NATURAL JOIN

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Have a nice day!